Transportation of ovum and fertilization

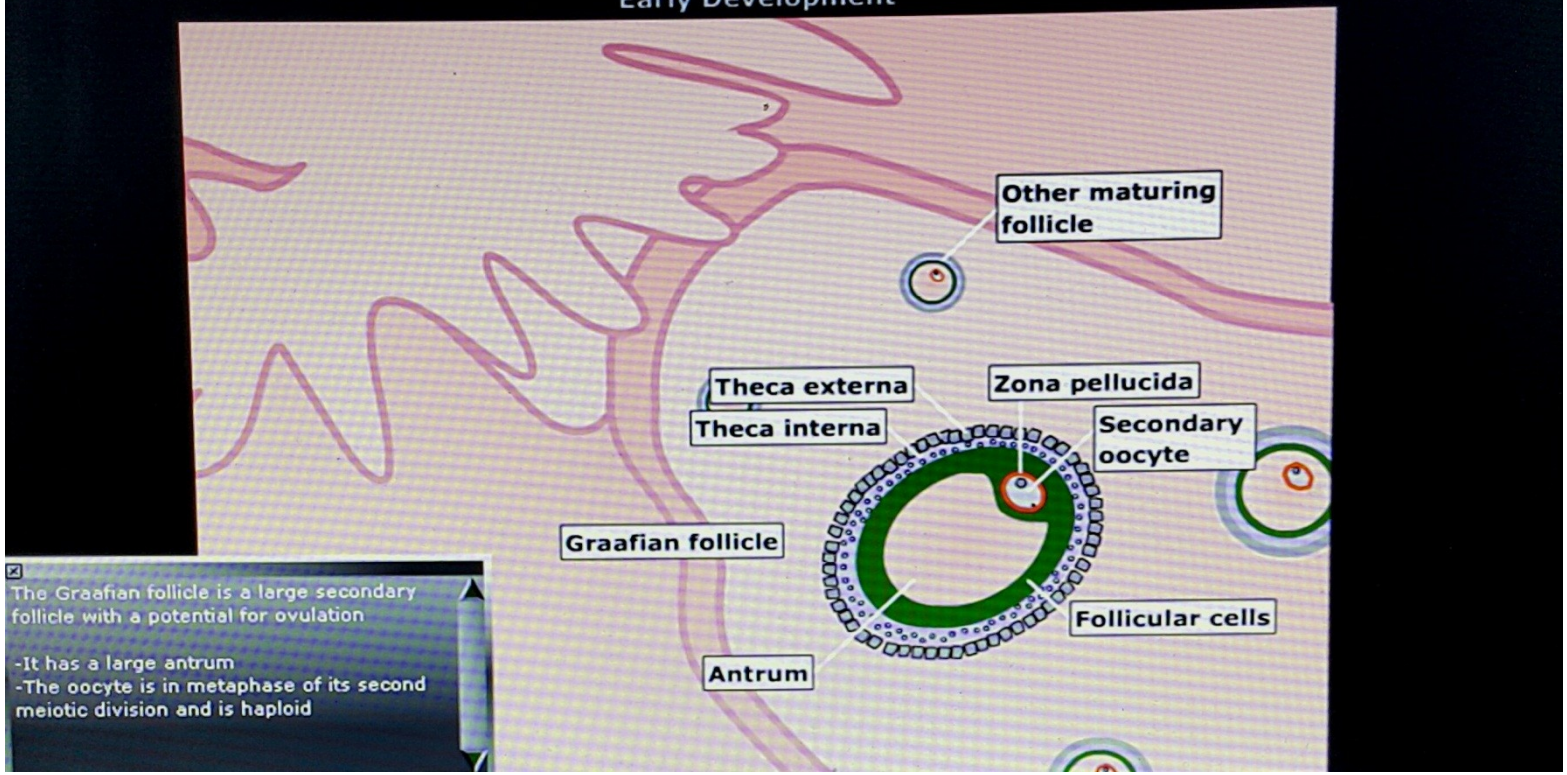
Learning Objectives

- **At the end of the lecture, the students should be able to :**
  - Discuss the transport of ovum from the surface of ovary to ampulla of fallopian tube
  - List the factors affecting the transport of ovum
  - Define fertilization
  - State normal site of fertilization
  - Describe the results of fertilization
  - Mentions the factors affecting fertilization
  - Enumerate the changes that occur in spermatozoa before fertilization
  - Explain the factors affecting penetration of sperm through the zona pellucida for formation of Pro nuclei

Maturation of Ovum

Day 0

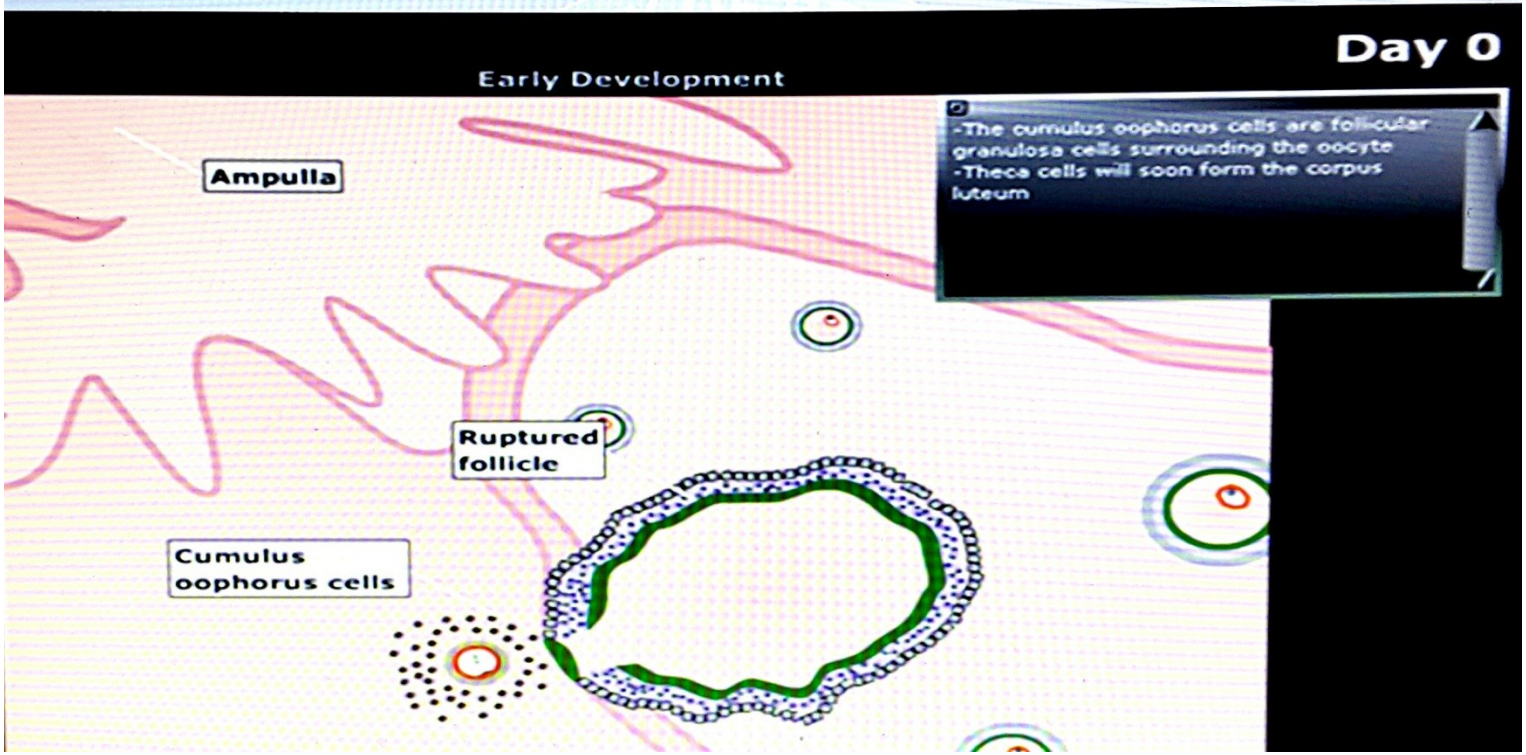
Early Development



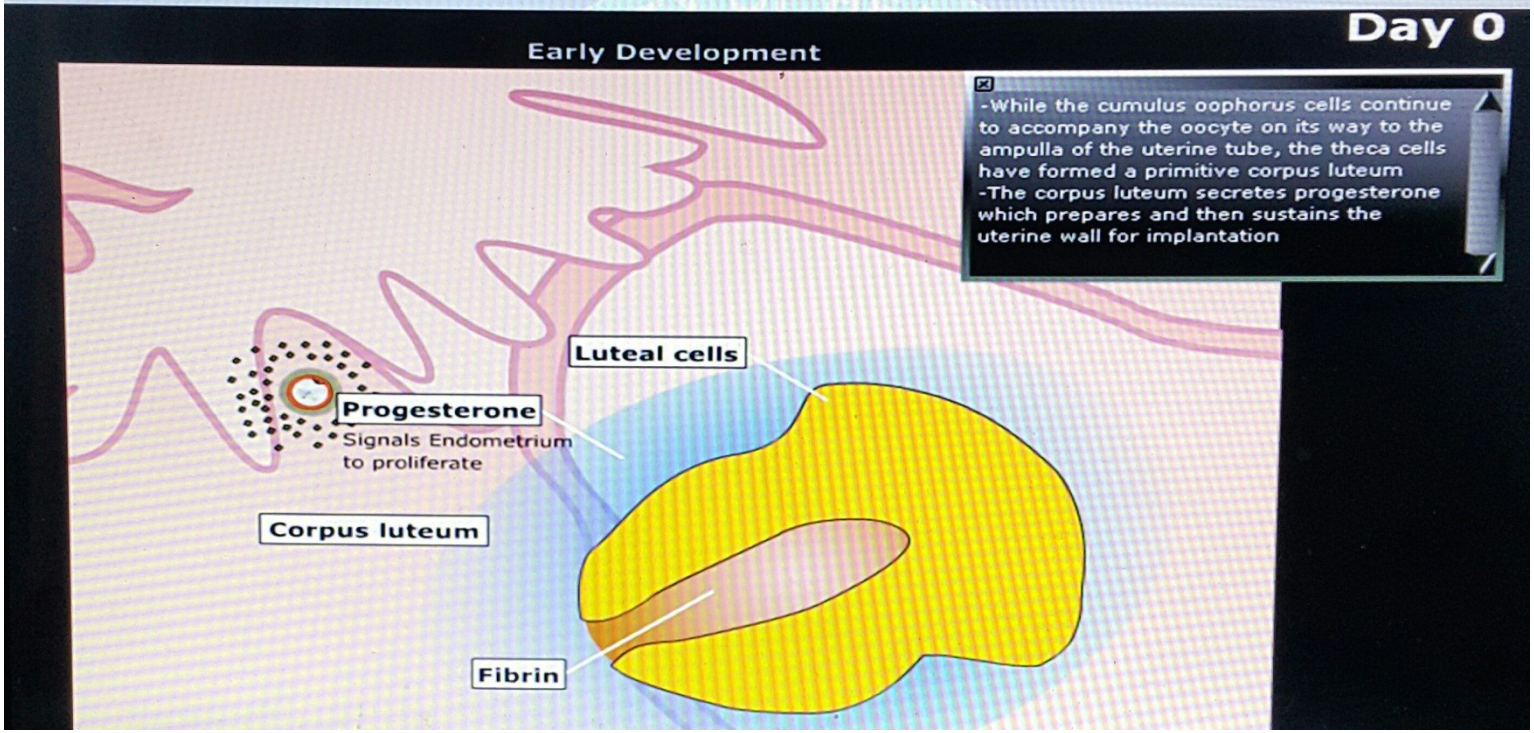
The Graafian follicle is a large secondary follicle with a potential for ovulation

-It has a large antrum
-The oocyte is in metaphase of its second meiotic division and is haploid

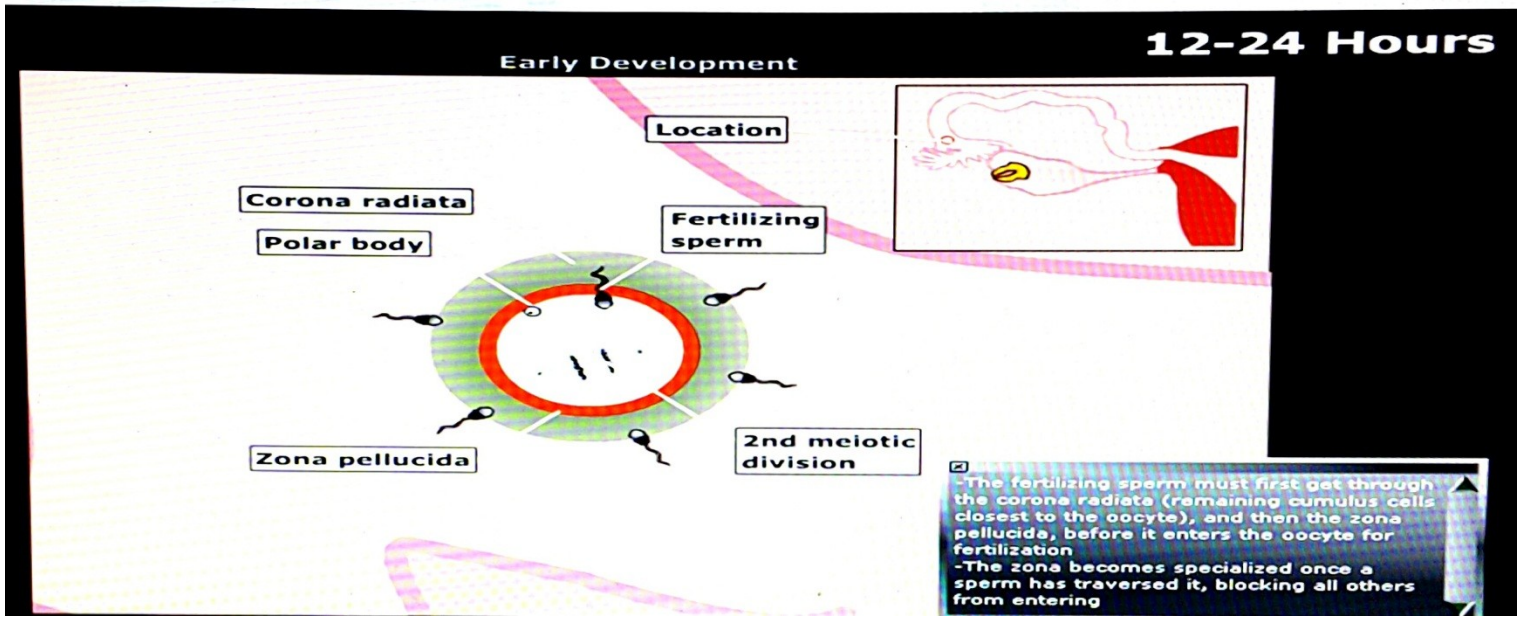
Release of ovum from ovary into abdominal cavity



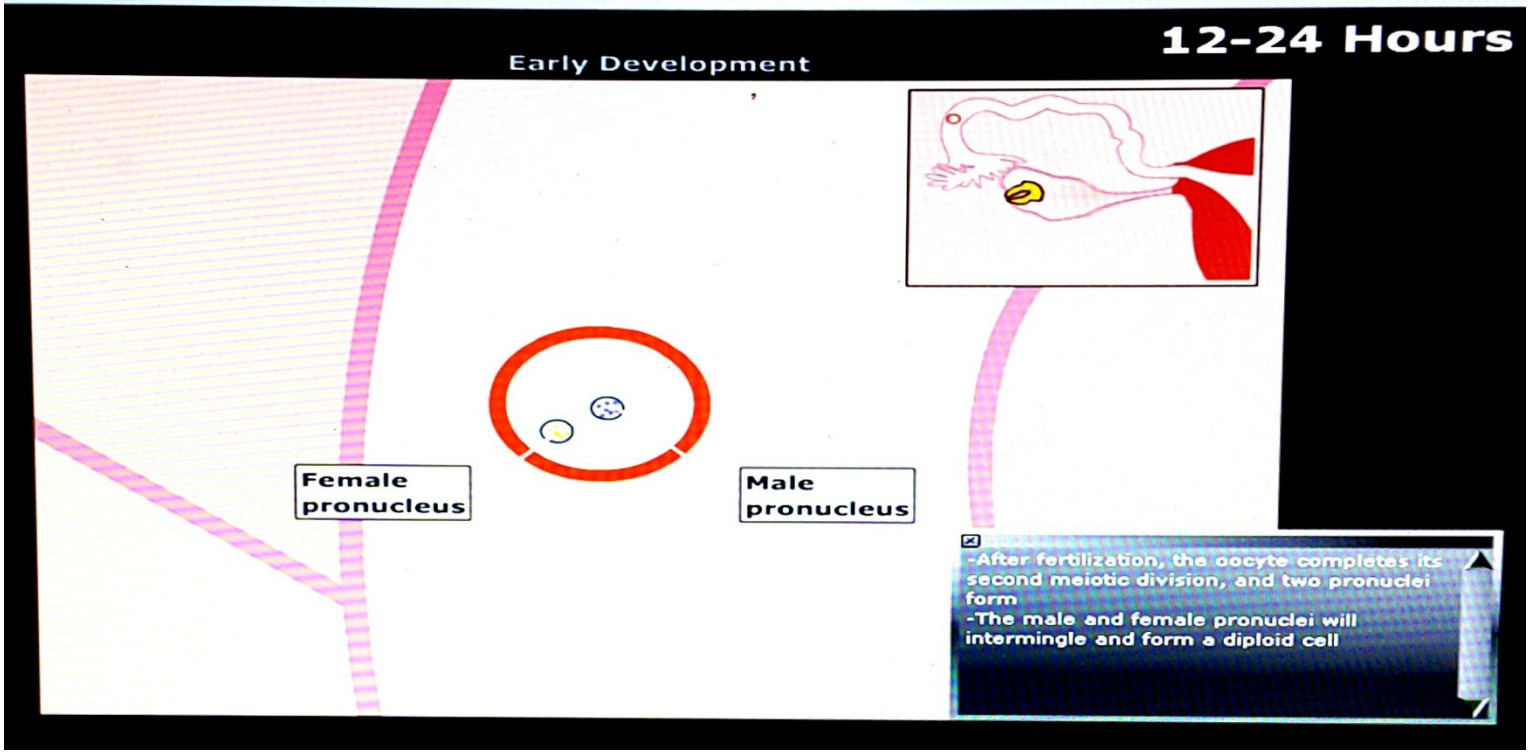
Formation of Corpus Luteum, Ovum is taken up by the fimbria and enters into first part of Fallopian tube that is Infundibulum



Ovum travels with in the fallopian tube with the cillia of tall columnar epithelium of mucosa and reaches the next dilated part of tube, Ampulla, where the ovum stays for 24 to 48 hours and the part where fertilization takes place



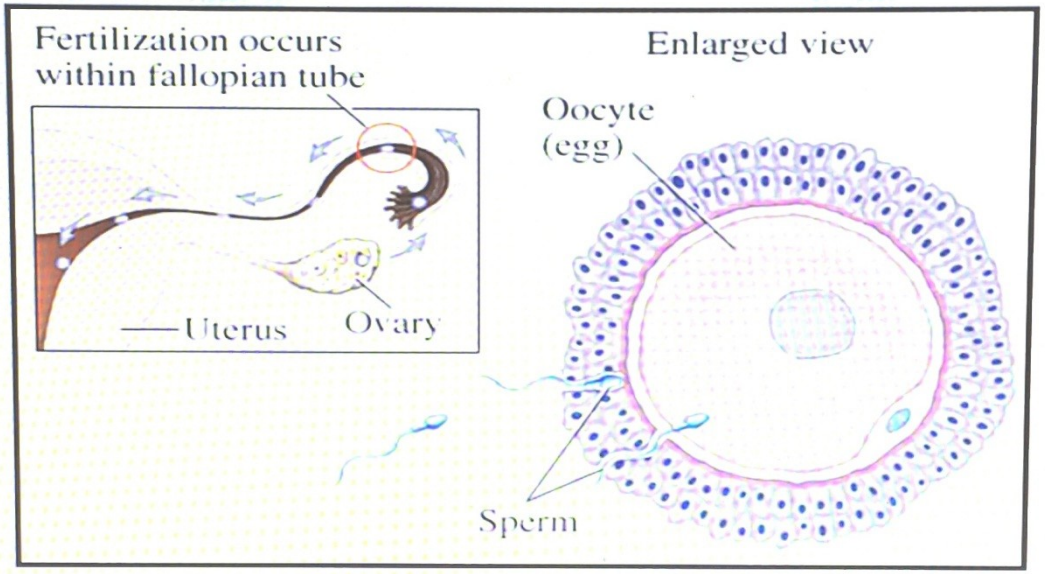
Formation of male and female Pro-nuclei



FERTILIZATION

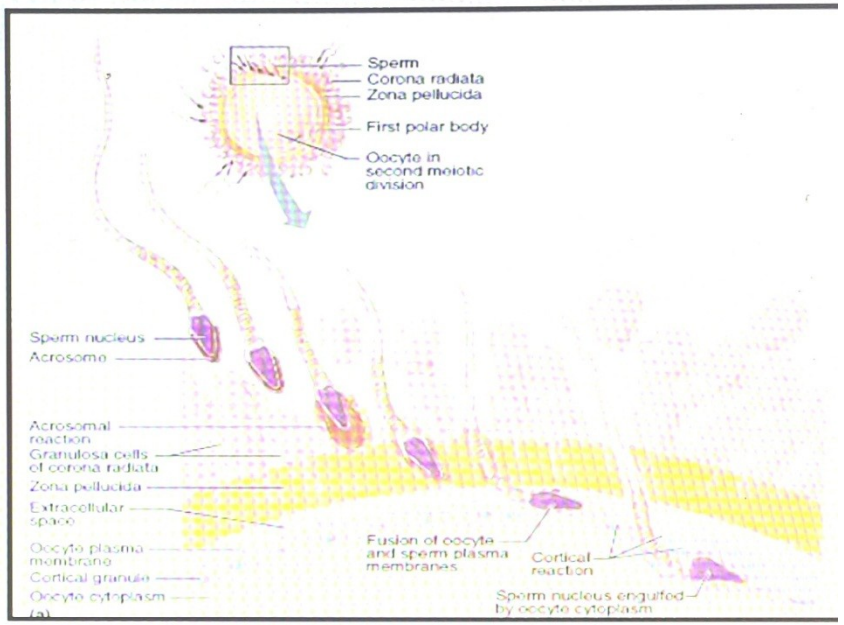
Definition:

The **union** of **male gamete (sperm,haploid cell)** and **female gamete (oocyte,haploid cell)** gamete to form a zygote(diploid cell).



FERTILIZATION

- Begins with a **contact** between the sperm & the ovum.
- Ends up with restoration of 46 chromosomes (diploid) each set of 23 chromosomes from the maternal and paternal

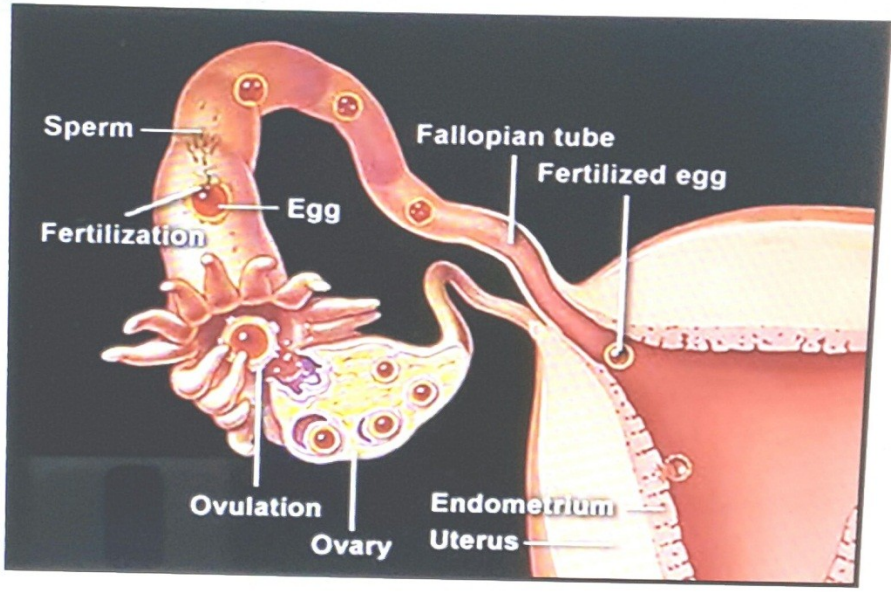


SITE OF FERTILIZATION

Usual site is the **ampulla of the uterine tube.**

Ampulla is the longest and widest part.

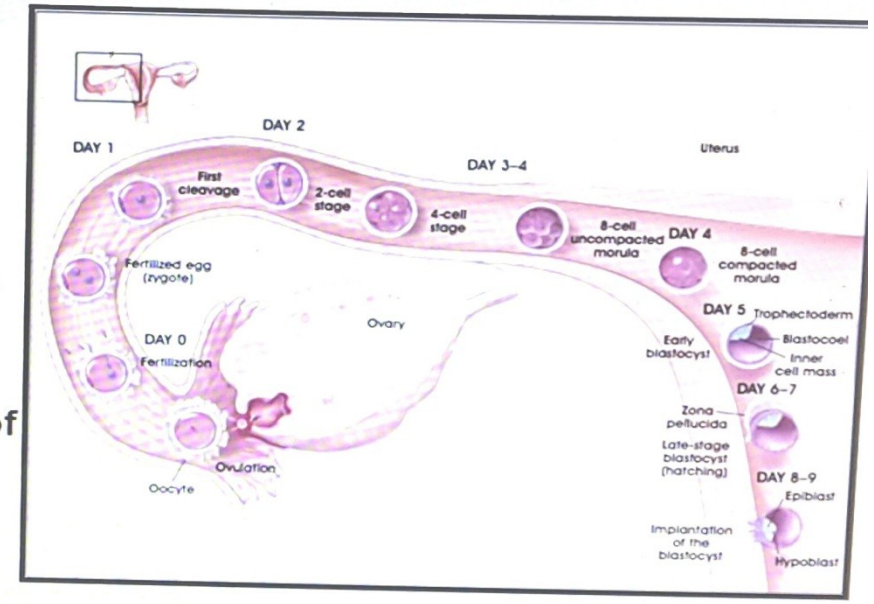
Chemical signals (attractants) , secreted by oocyte and surrounding follicular cells, guide capacitated sperms (sperm chemotaxis) to the oocyte.



PHASES OF FERTILIZATION

Fertilization is a sequence of coordinated events:

- Passage of a sperm through the corona radiata.
- Penetration of the zona pellucida.
- Fusion of plasma membranes of the oocyte and sperm.
- Completion of the second meiotic division & formation of female pronucleus
- Formation of the male pronucleus.
- Ootid Becomes a zygote..

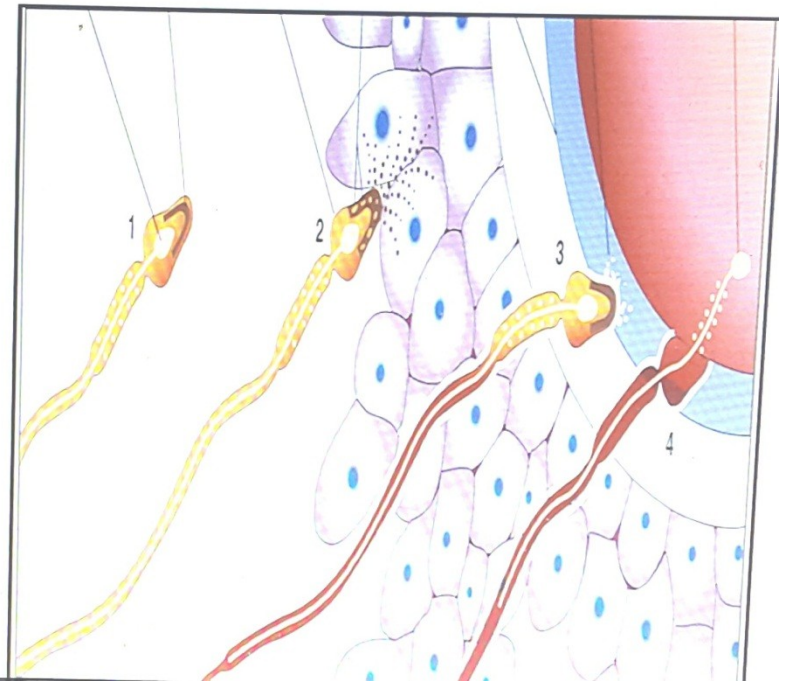


PHASES OF FERTILIZATION

1- Passage of sperm through corona radiata,

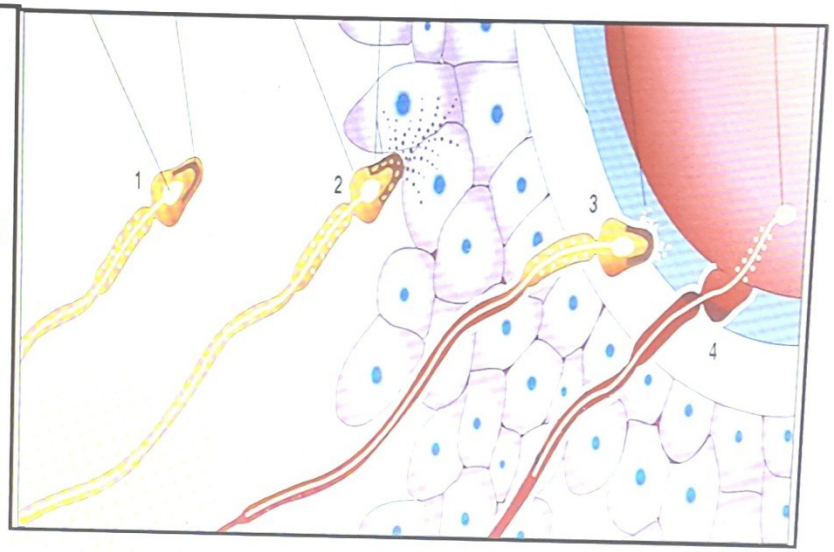
Under the effect of:

- Hyaluronidase enzyme from sperms,
- Tubal enzyme. And
- Movement of tail of sperm.



PHASES OF FERTILIZATION

- **2- Penetration of the zona pellucida by head of sperms through acrosine E. from acrosome of one sperm.**
- **3- Fusion of the plasma membrane of the oocyte and that of the sperm.**



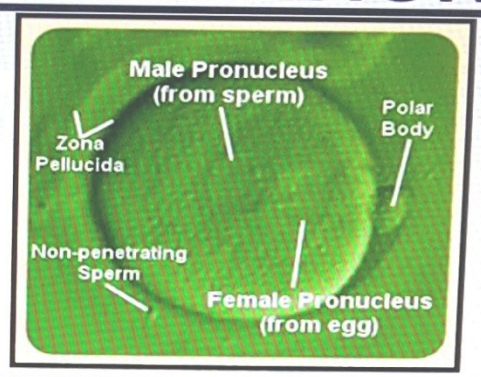
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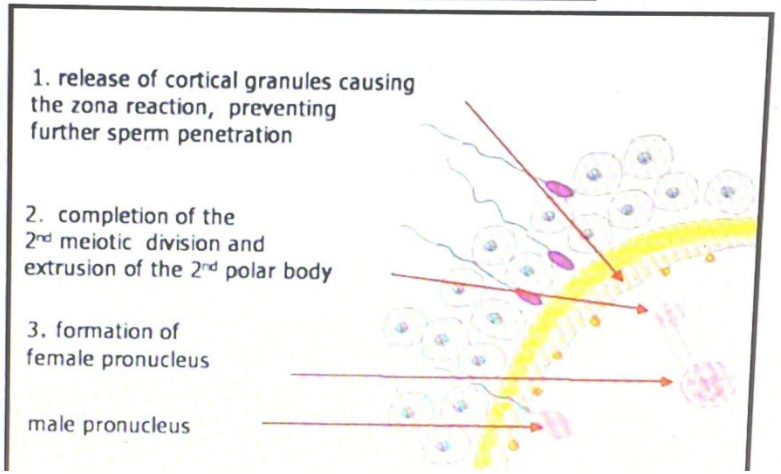
4- <u>Completion</u> of the second meiotic division & formation of the female pronucleus.

5- Formation of the male pronucleus :

It is a swollen nucleus of the sperm.

Its tail is detached and degenerated.





ZONA REACTION

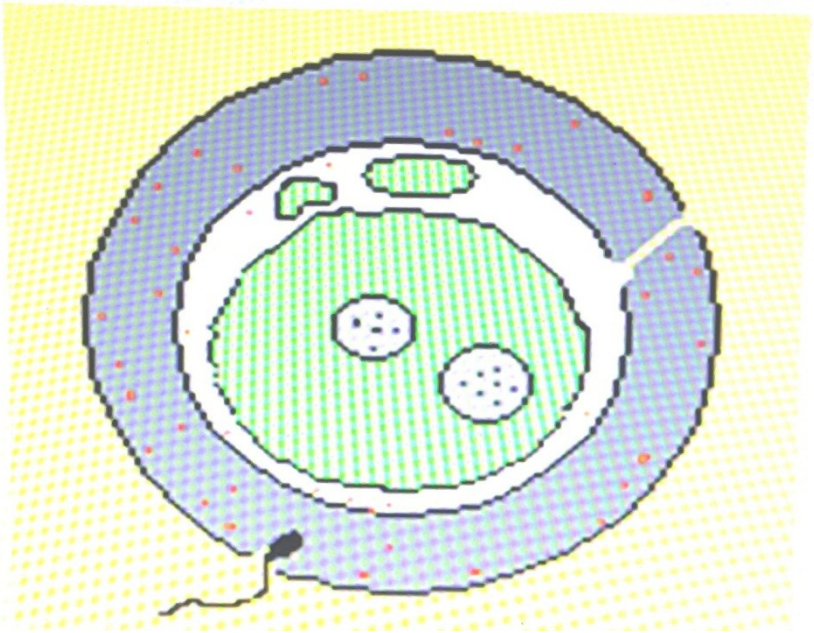
It is a change in properties of zona pellucida that makes it impermeable to other sperms.

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SEX OF THE EMBRYO

- Embryos chromosomal sex is determined at fertilization by the kind of sperm (X or Y) that fertilizes the oocyte.
- Fertilization by an X – bearing sperm produces a 46 , XX zygote, which normally develops into a female, whereas fertilization by a Y – bearing sperm produces a 46, XY zygote , which normally develops into a male.

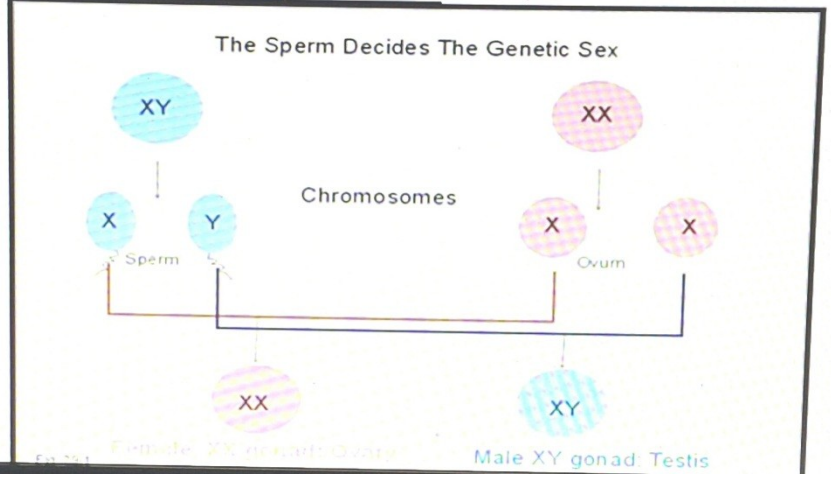
CHROMOSOMES IN ZYGOTE

Zygote is genetically unique.

Half of its chromosomes come from the father and the other half comes from the mother.

zygote contains 46 chromosomes (diploid).

This mechanism forms biparental inheritance and leads to variation of the human species.



RESULTS OF FERTILIZATION

- **Stimulates the penetrated oocyte to complete its 2nd meiotic division.**
- Restores the normal diploid number of chromosomes in the zygote (46).
- Variation of human species through mingling of maternal and paternal chromosomes.
- Determines the chromosomal sex of the embryo.
- Initiates cleavage (cell division) of the zygote.

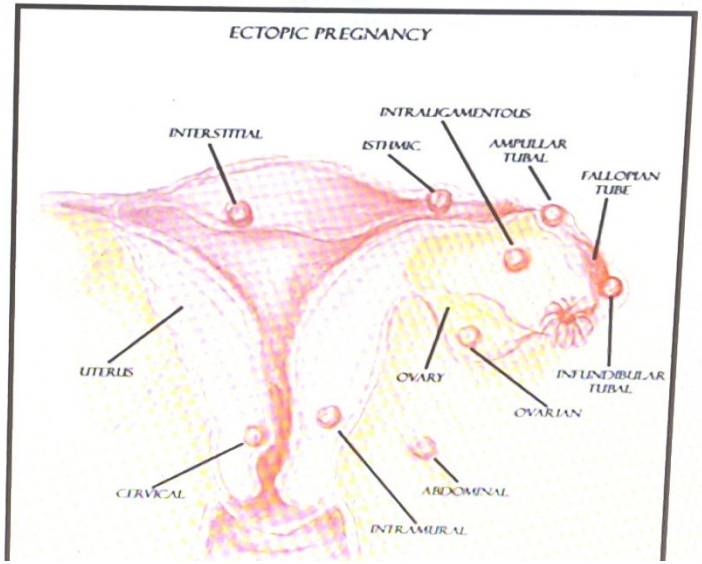
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ECTOPIC PREGNANCY

- **It means implantation outside the uterus.**
- 95 to 97% of ectopic pregnancies occurs in the uterine tube.
- Most are in the ampulla & isthmus.
- **Placenta previa :**
- Implantation occurs in the lower uterine segment.

- **Ectopic Pregnancy:**
  - **1- Placenta Previa.**
  - **2- Tubal.**
  - **3- Ovarian.**
  - **4- Abdominal.**
  - **5- Pelvic.**
  - **6- Cervical.**



ZONA REACTION

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References

- The Developing Human Clinical Oriented Embryology.

Keith L. Moore.

Eighth Edition.

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